

Patterns Without Source: A Meta-Analysis of PWS Cup 2025

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Abstract. Practical evaluation of data anonymization requires evidence from real-world attacks, not only formal privacy guarantees. Data anonymization competitions provide such evidence, but participants rarely release their code, making it a methodological challenge to reconstruct strategies from public presentations.

Using the PWS Cup 2025 health-data anonymization competition as a case study, we reconstructed the strategies of 20 teams and encoded them as categorical features. We then related these features to competition scores using effect sizes and parsimonious models. Our analysis examines two hypotheses: generation- or reconstruction-based designs are associated with higher utility and anonymization scores than direct perturbation-based approaches in this contest setting, and stronger anonymization reduces disclosure risk across attack channels. We interpret these patterns as exploratory evidence and conclude by discussing what information future competitions should provide to enable more rigorous evaluation.

Keywords: Meta-analysis · Synthetic data · Anonymized data · Data anonymization competition

1 Introduction

Data anonymization and data synthesis are key technologies for sharing sensitive data while reducing the risk of revealing individual information [16]. They are particularly important for health data, where secondary analysis is beneficial, and disclosure can be harmful.

Evaluating whether anonymization truly protects individuals requires more than just formal guarantees; it demands practical red-team testing, in which adversaries actively attempt to re-identify records from anonymized or synthetic data. This perspective has gained wide acceptance [6], and competitions have emerged as a prominent venue for such evaluation. For example, the MIDST challenge at SaTML 2025 [12] drew 71 participants to conduct membership inference

attacks against diffusion-model-based synthetic tabular data, while the SNAKE challenge [1] invited teams to attack differentially private synthetic data generators. Both competitions mandated the release of open-source code as a condition for winning prizes: MIDST required award-eligible participants to publish their code⁵, and the SNAKE challenge similarly required top-placing teams to release their attack algorithms under an open-source license⁶. Their code disclosure policy is expected to enhance our understanding of anonymization technologies and their vulnerabilities.

A fundamental challenge in the real world is that attacks and defenses often occur without mutual disclosure. Organizations that use anonymization pipelines seldom share their implementation details, while adversaries—whether they are malicious actors or authorized auditors—also keep their methods hidden. In many operational settings, we can observe the outcomes, but the algorithms that generated those outcomes remain unclear. How can we draw lessons from situations where neither the attack nor the defense codes are disclosed?

This paper addresses the question through a meta-analysis of the PWS Cup 2025. The PWS Cup, held annually in Japan since 2015 [9, 10], requires teams to both anonymize data and attack the anonymized outputs of others. Although code disclosure is optional and only two teams have released their code, all teams present their approaches at the final showcase. We use these presentations as the basis for our analysis.

We examined the strategies of 20 out of the 24 teams based on the posters and short presentations showcased during the competition. Each strategy was encoded using a set of categorical features, and we related these features to measures of utility, anonymization, and attack scores. Throughout this process, we adhered to the study’s constraints to guide our methodology. We focused on effect sizes rather than relying solely on significance tests. We also cross-verified univariate results with regression analysis that used AICc [2, 8] to select parsimonious models, and we presented all our empirical findings.

Our contributions are summarized as follows.

- We frame the meta-analysis of a data anonymization competition as a problem in its own right, and articulate the constraints – a small sample and undisclosed participant code – that any such analysis must confront.
- We propose a methodology for attempting the analysis responsibly under those constraints: a manual reconstruction and categorical coding of each team’s strategy, and a small-sample-robust analysis that foregrounds effect sizes and cross-checks them against AICc-selected parsimonious regression.
- We demonstrate the methodology on the PWS Cup 2025, reporting the strategies actually used and the hypotheses the analysis surfaces about how those strategies relate to the utility, anonymization, and attack scores. The analysis is organized around two concrete hypotheses.
- We are explicit about what the constraints preclude: we report the empirical results as exploratory hypotheses rather than confirmatory conclusions, and

⁵ See Terms tab in <https://www.codabench.org/competitions/4671/>

⁶ See Terms tab in <https://www.codabench.org/competitions/879/>

discuss what a future competition would need to provide for them to be tested.

2 PWS Cup 2025

2.1 Competition overview

Of the 24 registered teams, 18 disclosed their affiliations: 13 were primarily student teams, and 5 were from industry. Most participants, therefore, appeared to have backgrounds in computer science and information security rather than privacy-enhancing technologies. The competition uses healthcare datasets generated by Synthea [17]. The dataset contains 18 variables: three categorical variables (gender, race, and ethnicity) and fifteen numerical variables, including age, diagnosis flags, blood pressure, BMI, weight, and medical-utilization measures.

The competition consisted of two phases. In the anonymization phase shown in Fig. 1(a), the secretariat generates a population dataset A_i (10^5 rows, 18 columns) using Synthea and provides each team i with a sample B_i of 10^4 rows drawn from A_i . Each team produces an anonymized dataset C_i from B_i and trains an XGBoost model D_i [4] that predicts a patient’s stroke risk, i.e., the binary attribute `stroke_flag`, from the remaining attributes.

The anonymization score rewards teams whose data resists membership inference. Let MIA_i be the largest number of correct membership inferences that any attacker achieves against B_i , out of 10,000; that is, $\text{Anon_score} = (10000 - MIA_i)/100$. To protect every patient rather than the average one, a penalty is also applied: if some record of B_i is correctly identified by L distinct attackers, then $\text{Anon_pen} = L - 1$ for the largest such L . The total anonymization score $\text{Anon_all} = \max\left(0, (10,000 - MIA_i)/100 - \text{Anon_pen}\right)$.

The utility score consists of four components. U_1 measures the preservation of statistical properties between the original dataset B_i and the anonymized dataset C_i . U_2 evaluates the consistency of logistic regression results. U_3 assesses the preservation of group-comparison results based on Kruskal–Wallis tests. U_4 measures the predictive accuracy of the submitted XGBoost model D_i . The overall utility score Utility_all is defined as

$$\text{Utility_all} = U_1 + U_2 + U_3 + U_4,$$

with a maximum score of 100 points.

The score of team i in a round is the sum of its utility and anonymization scores; we write $s_{p,i}$ and $s_{m,i}$ for the preliminary- and main-round scores. The final score is their weighted sum, $s_i = 0.1 s_{p,i} + 0.9 s_{m,i}$.

In the attack phase shown in Fig. 1(b), each team i receives the population and anonymized datasets, A_j and C_j , of every other team j . Using C_j and the corresponding model D_j , team i performs a membership inference attack [13] to determine whether records in A_j belong to the undisclosed sample B_j . This process is repeated for all other teams. The attack score Attack_t10_score is

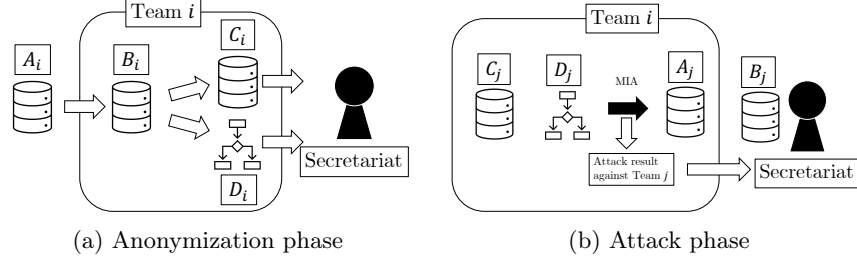


Fig. 1: System model of PWS Cup 2025

the number of correct inferences against the ten highest-ranked teams, divided by 100, while `Attack_sum_score` is the total number of correct inferences across all teams. Table 1 shows the main results of PWS Cup 2025. Please refer to the official website for more details about PWS Cup 2025⁷.

2.2 Strategies of participating teams

At the end of the competition, the final scores and score breakdowns were published, but the algorithms used by each team were not disclosed. Participants were required to submit only anonymized data, XGBoost models, and attack results, and the final showcase provided only high-level descriptions of their approaches. This limited disclosure makes it difficult to extract broader insights from the competition.

To recover the missing information, we attended the final showcase and documented each team’s anonymization, model-training, and attack strategies. We obtained usable summaries for 20 of the 24 teams, as two teams partially or completely withdrew from the competition. The anonymization and attack strategies reconstructed from these summaries are described below.

Anonymization Strategies We observed two broad approaches for generating the anonymized dataset C_i . The first generates a new dataset that mimics the statistical properties of B_i . Examples include Synthea-based synthesis (teams 10 and 18), Gaussian copulas [11] (teams 2 and 13), and CTGAN [18] (team 16). Many teams then refined the synthetic data to improve utility using optimization techniques such as hill-climbing, simulated annealing, or genetic algorithms.

The second approach transforms B_i to C_i directly, through data masking methods for k -anonymization [15] such as noise addition, generalization, and record swapping (e.g., teams 9, 22, and 23). Some teams, including ours, tried data synthesizers designed for differential privacy [5], but they failed to achieve good utility and abandoned them; only two teams used explicit Laplace noise.

⁷ <https://www.iwsec.org/pws/2025/cup25.html>, which is in Japanese

The training D_i strategies are not so varied. One remarkable trend is that some teams did not use B_i for D_i training. Because U_4 is a test accuracy, fresh data generated by Synthea is more useful than B_i for those teams.

Attack Strategies Many teams conducted attacks based on record-level similarity; that is, they matched records of the population A_j to those of the anonymized data C_j under a distance metric such as Manhattan, Euclidean, Gower, or Mahalanobis distance, often resolving the matches into a one-to-one assignment using the Hungarian algorithm. A small number of groups used model-based membership attacks rather than relying only on direct record-to-record distances. These included random-forest membership classifiers, likelihood-ratio attacks such as LiRA [3] and RMIA [19], and graph- or SHAP-based scoring methods. Several teams ran multiple attacks and kept the best result for each target, and many fell back to random guessing against strongly anonymized teams. The strongest attacker, team 8, combined a Gaussian LiRA with a distance attack on covariance-augmented data.

3 Meta-Analysis Methodology

We present a meta-analysis methodology based on categorical features representing the design choices of each team. Using these features, we develop statistical models to predict team scores and quantify the impact of individual design choices on competition outcomes.

3.1 Motivational Hypotheses for Feature Coding

Two hypotheses emerged from the presentations at the competition showcase. The most salient contrast, reflecting the long-standing debate over whether synthetic data generation offers a genuine advantage over traditional perturbation-based approaches [14], was between teams that generated a new dataset from sources other than B_i and those that directly perturbed B_i . This observation motivated our first research question.

H1: Are generation-based approaches associated with better utility-anonymization outcomes than direct perturbation-based approaches?

On the attack side, two dimensions of variation stood out: whether a team targeted the submitted model D_j alongside C_j , and whether it used model-based membership attacks, in which the attacker fitted its own scoring rule, rather than record-level distance matching. These motivated our second question.

H2: Does the combination of a victim’s anonymization strategy and an attacker’s attack strategy jointly determine disclosure risk in the membership inference game?

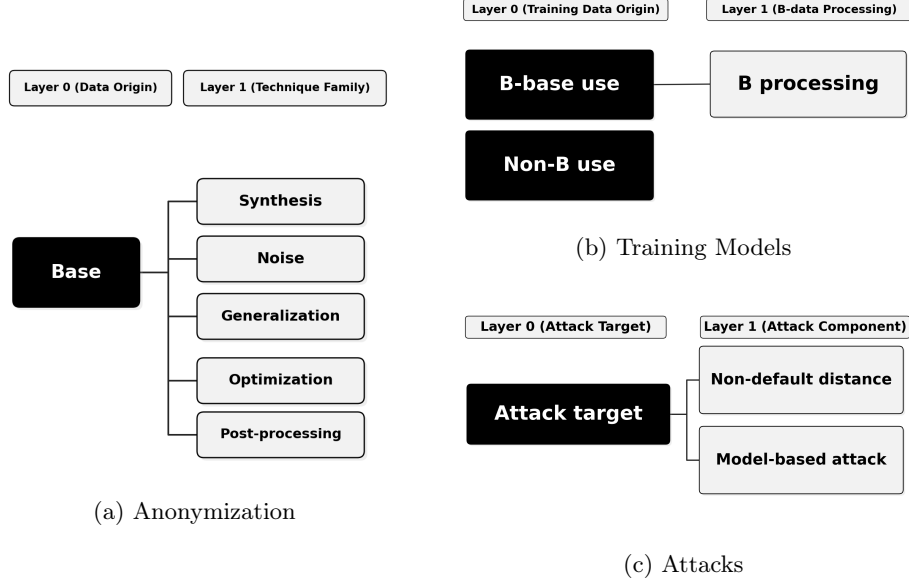


Fig. 2: Taxonomies of Participating Teams' Strategies

3.2 Defining Features

We define categorical features relevant to evaluating **H1** and **H2**. To keep the analysis tractable with a sample size of $n = 20$, we retained only features that were present in at least four teams, ensuring that any observed effect was not driven by a few idiosyncratic teams.

The features are defined hierarchically, as illustrated in Fig. 2. Specifically, we group the features into three categories—Anonymization, Training Models, and Attacks—and construct a feature hierarchy for each category. Features at the top level represent high-level design choices, whereas those at lower levels capture progressively more specific refinements of their parent features.

We describe features in each group below.

Anonymization features. Variable **Base** = {**B**, **others**} encodes top-level feature for anonymization strategies. **B** denotes transform-based anonymization from B_i to C_i , whereas **others** generates C_i without using B_i . When **Base** = **others**, variable **Synthesis** = {**yes**, **no**} specifies whether C_i is synthetically generated or not. When **Base** = **B**, variable (**Noise** = {**yes**, **no**}) specifies whether the method of noise addition is used or not.

There are several orthogonal design features that may influence the utility and privacy of C_i . Variable **Optimization** = {**yes**, **no**} indicates whether an explicit optimization procedure is employed to improve the utility of C_i , particularly for synthetic-data generation methods. Variable **Post-processing** = {**none**, **swap**} indicates whether additional privacy-enhancing transformations,

such as row swapping and permutation, are applied on top of the core anonymization strategy.

Training features. Training features in Fig. 2b characterize strategies for training an XGBoost model for stroke-risk prediction (see Section 2). Variable **B-base use** = {yes,no} specifies whether the training process uses B_i -based data to train the model. Variable **Non-B use** = {yes,no} specifies whether the training process uses another dataset, which is different from B_i . Variable **B processing** = {none,synthetic,NA} specifies how B_i -based data is processed before training the model on the condition that **B-base use** = yes. **B processing** = none means B_i is used as-is; **B processing** = synthetic means synthetic data generated from B_i is used instead; **B processing** = NA when **B-base use** = no.

Attack features. The attack score measures how successfully a team identifies records used as anonymization targets (see Section 2); the following features, illustrated in Fig. 2c, characterize each team’s attack strategies.

Variable **Attack target** = {anon,both} specifies whether an attack only considers anonymized data C_j , or also exploits the prediction model D_j . Variable **Model-based attack** = {yes,no} specifies whether an attacker conducts a membership inference attack based on a trained model or uses an alternate method based on the distance of matching records. Variable **Non default distance** = {yes,no} specifies whether the record-level component uses any distance metric besides the baseline Manhattan distance.

We summarize the breakdown of the number of teams for each feature value in Table 2 in Appendix A.

3.3 Testing H1: Score Regression Model

To verify **H1**, we model each competition score as a linear function of the teams’ strategy features in Section 3.2. Let y_i denote a competition score for team i and let $f_{1i}, \dots, f_{pi}$ be its strategy features where f_{ki} is the team i ’s k -th feature. We model:

$$y_i = \beta_0 + \sum_{k=1}^K \beta_k f_{ki} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (1)$$

where each β_k measures the expected change in the score associated with feature f_k , all else equal. A separate instance of (1) is fitted for each score (**Utility_all**, **Anon_score**, **Anon_all**, **Anon_pen**, **Anon_sum_score**, **Attack_t10_score**, **Attack_sum_score**), where **Anon_sum_score** is the mean of $(10,000 - r_{ji})/100$ across all valid attackers i , and **Attack_sum_score** is the analogous attack score aggregated over all target teams rather than only the top ten. Because the sample is small ($n = 20$), we restrict to $K \leq 3$ main effects and at most one two-way interaction.

Univariate screening. For every (score, feature) pair, we split the 20 teams by the feature and compute an effect-size summary. For binary features, we use

Hedges’ g [7], the small-sample-corrected standardized mean difference:

$$g = J(n_1 + n_2 - 2) \frac{\bar{y}_1 - \bar{y}_2}{S_{\text{pooled}}}, \quad J(m) = 1 - \frac{3}{4m - 1}, \quad (2)$$

Here, groups 1 and 2 denote the two categories of the feature being compared; n_1 and n_2 are their sample sizes, and $\bar{y}_1$ and $\bar{y}_2$ are their mean scores. S_{pooled} is the pooled standard deviation across the two groups, $m = n_1 + n_2 - 2$ is the degrees of freedom, and $J(m)$ is the bias-correction factor for small samples. We follow the convention that $|g| > 0.8$ is a large effect. Univariate screening identifies which features are most strongly associated with each score and guides feature selection for (1).

The regression analysis serves as an exploratory complement to the univariate effect-size analysis. While Hedges’ g measures how strongly each feature separates the observed scores, large univariate effects may reflect confounding or correlations with other features. Regression models, therefore, assess which features retain their associations when considered jointly.

Due to the small number of teams, we restrict the analysis to sparse models with at most three main effects and one interaction term. Models are selected using the corrected Akaike information criterion (AICc) [2, 8], which penalizes complexity and is appropriate for small samples.

We do not use these regressions as predictive models and do not claim out-of-sample predictive validity. Their role is diagnostic: they test whether the main univariate patterns remain visible after controlling for a small number of co-coded strategy features. A full validation study would require either substantially more competitions or simulation data with known data-generating mechanisms. We therefore treat the selected regressions as robustness checks for exploratory patterns, not as confirmatory causal models.

3.4 Testing H2: Dyadic Disclosure Model

To test **H2**, team-level scores are insufficient because they obscure the compatibility between anonymization and attack strategies. We therefore analyze the attack-result matrix directly, constructing one observation for each ordered victim–attacker pair. This dyadic structure preserves the membership inference game, in which each victim is attacked by multiple attackers, and each attacker targets multiple victims.

The response r_{ji} is `Exposed_Records`, the count of victim j ’s records that attacker i correctly identified. This outcome is a nonnegative count rather than a continuous score, and it varies strongly across victim–attacker pairs. Because the count data exhibit substantial over-dispersion, we use a negative binomial regression model, which retains a log-linear specification while allowing the variance to exceed the mean:

$$\log \mathbb{E}[r_{ji}] = \alpha + \beta_a^\top \mathbf{a}_j + \beta_c^\top \mathbf{c}_i + \beta_{ac}^\top (\mathbf{a}_j \otimes \mathbf{c}_i) \quad (3)$$

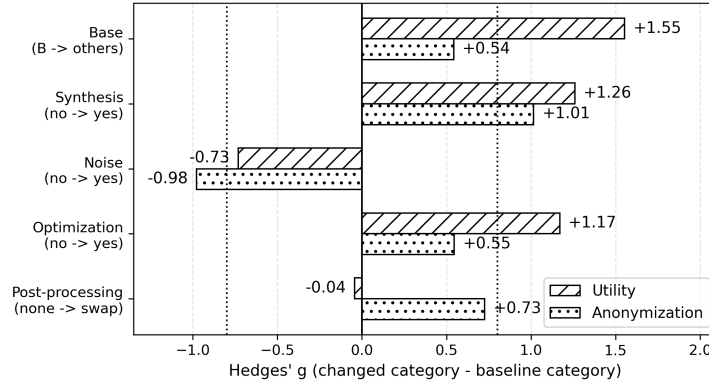


Fig. 3: Hedges’ g for each strategy feature on utility (`Utility_all`) and anonymization (`Anon_all`). Each contrast is oriented in the direction shown on the y-axis; positive values indicate higher scores for the right-hand category. Dotted lines mark $|g| = 0.8$ (large effect).

where $\mathbf{a}_j$ is a victim-side anonymization feature and $\mathbf{c}_i$ is an attacker-side attack feature. Standard errors are two-way clustered by attacker and victim to account for repeated measurements arising from the same attack procedure across multiple victims and from repeated attacks against the same victim. Exponentiating a coefficient yields an incidence rate ratio (IRR), which represents the multiplicative change in the expected number of exposed records; an IRR below 1 indicates reduced disclosure risk. The model is fitted separately for each of the eight combinations of victim-side and attacker-side features, formed by the four victim-side anonymization features and the two attacker-side attack features. The analysis is based on the 361 ordered attacker–victim pairs remaining after excluding self-pairs and teams with missing data.

4 Meta-analysis Results

We evaluate the validity of the two hypotheses in Section 3.1 using meta-analysis.

4.1 H1: Utility and Anonymization

Hypothesis **H1** in Section 3.1 asks whether generation- or reconstruction-based strategies are associated with better utility–anonymization outcomes than direct perturbation-based anonymization in PWS Cup 2025. We examine each outcome separately. Figure 3 summarizes the univariate effect sizes of the strategy features across both outcome dimensions.

Three strategy features show large and consistent associations with the utility scores. The base of the anonymization design, `Base`, has the strongest effect:

teams that did not start from the given sample B_i (**Base** = **others**) outscored teams that did (**Base** = **B**) on every utility component, with Hedges' $g = 1.55$ for the total utility score **Utility_all**. Whether the design was synthetic, **Synthesis**, is the strongest single driver of the logistic-regression component U_2 ($g = 1.86$, favoring **synthetic**) and is large for total utility ($g = 1.26$). **Noise** has the weakly negative effect on the utility ($g = -0.73$ for **Utility_all**). The presence of an explicit optimization step, **Optimization**, is similarly large ($g = 1.17$ for U). The additive technique (swapping), in contrast, has essentially no effect on utility ($|g| < 0.25$ throughout). Model selection by AICc chose **Base** alone for U_1 and U_3 , **Synthesis** alone for U_2 .

Although **Optimization** exhibits a large univariate effect, it was not retained in any selected model. This likely reflects its overlap with broader design features: four of the six teams using optimization also relied on a non- B_i base, and optimization was typically associated with synthetic or reconstruction-based approaches. Consequently, once **Base** or **Synthesis** is included in the model, **Optimization** provides insufficient additional explanatory value to justify its inclusion under the AICc criterion.

Overall, the highest utility was achieved not by lightly perturbing the provided sample but by reconstructing a fresh synthetic dataset, typically via an optimization loop, that reproduces the target statistics.

The anonymization scores are governed by a different set of features. **Base**, the dominant driver of utility, has only a small effect on the base anonymization score (**Anon_score**, $g = 0.48$). The largest positive effects instead come from **Synthesis** ($g = 1.01$ for the post-penalty score **Anon_all**; synthetic designs were safer), the additive technique ($g = 0.73$; swapping improved the score), and, among the noise-using teams, the noise target ($g = 0.90$; applying noise to both numeric and categorical attributes rather than to numeric attributes only). The penalty term **Anon_pen** is most associated with the noise type ($g = 0.84$; Laplace noise drew larger penalties) and again with swapping ($g = -0.59$; swapping lowered the penalty).

The selected regressions for the anonymization scores combine **Synthesis** and **Post-processing** with their interaction ($R^2 = 0.47\text{--}0.69$); the interaction is itself sizeable (e.g., $p = 0.013$ for **Anon_sum_score**). It indicates that swapping helps most when added on top of an **others**-style design: in the fitted model, swapping raises the base anonymization score by about 32 points for **others** designs but leaves synthetic designs, already safe, essentially unchanged.

Overall, these results are consistent with **H1** in the specific setting of PWS Cup 2025. Both the univariate effect sizes in Fig. 3 and the AICc-selected multivariate regressions in Fig. 4 point in the same direction: generation- or reconstruction-based strategies were associated with higher utility and anonymization scores than direct perturbation-based strategies. Given the small sample size and the observational nature of the reconstructed features, we interpret this pattern as hypothesis-generating rather than confirmatory evidence. The observed

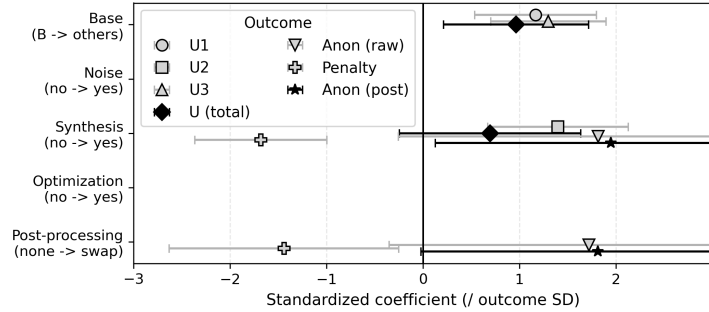


Fig. 4: Standardized OLS coefficients for the AICc-selected models, divided by the outcome SD. Each point is one (feature, outcome) pair; horizontal bars are 95% CI. Each contrast is oriented in the direction shown on the y-axis; positive values indicate higher scores for the right-hand category. A missing point-and-interval for a given outcome indicates that the feature was not retained in the AICc-selected model for that outcome variable.

outcomes are primarily driven by related but not identical sets of features: utility is mainly influenced by **Base** and **Synthesis**, while anonymization is associated with **Synthesis**, **Post-processing**, and their interactions. Notably, **Synthesis** is strongly associated with both utility and anonymization.

4.2 H2: Disclosure Risk and the Attack–Anonymization Interaction

H2 asks whether the combination of a victim’s anonymization strategy and an attacker’s attack strategy jointly determines disclosure risk. Before examining their interaction, we briefly characterize the two attack dimensions.

Attack strategies. Using a component in **Model-based attack** is associated with a higher top-10 attack score ($g = -0.86$, the model-based attack group scoring higher); the selected regression keeps this feature but explains little variance ($R^2 = 0.16$). Variable **Attack target**, points the other way: teams that attacked only the anonymized data scored higher than teams that also exploited the submitted model ($g = 0.70$ for **Attack_t10_score** and $g = 0.91$ for **Attack_sum_score**). In summary, applying an attack-side membership-scoring rule was somewhat helpful, whereas integrating the released model into the attack generally proved unbeneficial.

Figure 5 shows the full set of main and interaction effects from the dyadic model of Section 3.4 on the 361 ordered team pairs. The heat map summarizes the main effects and the interaction terms in the same IRR scale. The rounded rectangles in the heat map distinguish Area (a), the victim-side main effects; Area (b), the attacker-side main effects; and Area (c), the victim–attacker interaction effects. All four victim-side anonymization features reduce disclosure

Victim Strategy	Baseline	(b) Main: 0.82-0.86**	Main: 1.04-1.05
	Ref (1.00)		
	(a) Main: 0.72-0.73*	(c) Inter: 1.13** (Total: 0.69)	Inter: 1.03 (SE n.e.; Total: 0.78)
	Base (B -> others)		
	Synthesis (no -> yes)	Inter: 1.18** (Total: 0.61)	Inter: 1.00 (SE n.e.; Total: 0.68)
Post-processing (no -> yes)	Optimization (no -> yes)	Inter: 1.11** (Total: 0.70)	Inter: 1.01 (SE n.e.; Total: 0.78)
	Main: 0.68-0.70*	Inter: 1.16** (Total: 0.65)	Inter: 0.99 (SE n.e.; Total: 0.74)
	Baseline	Attack target (anon -> both)	Attacker-trained model attack (no -> yes)
	Attacker Strategy		

Fig. 5: Disclosure-risk effects from the dyadic negative binomial model ($n = 361$ ordered team pairs). Cells report incidence rate ratios (IRRs), where values below one indicate reduced risk. Areas (a)–(c) show victim effects, attacker effects, and their interactions. Stars denote statistical significance (* $p < 0.05$, ** $p < 0.01$).

risk substantially and consistently (IRR 0.63–0.73, all $p < 0.05$ under two-way clustering), corresponding to a 27–37% reduction in expected exposed records relative to a victim that used B_i without modification. On the attacker side, using the released model alongside the anonymized data (**Attack target** = both) is itself slightly disadvantageous as an attack (IRR 0.83–0.86, $p < 0.01$), while a model-based attack component (**Model-based attack**) shows no discernible effect on disclosure risk (IRR ≈ 1.04).

The four victim $\times$ **Attack target** interactions are all positive and significant under two-way clustering (IRR 1.10–1.16, $p \leq 0.003$): against a well-anonymized victim, the main-effect disadvantage of the model-based attack is partly canceled. The corresponding total effects nevertheless remain below one (IRR 0.63–0.69), so the victim-side anonymization advantage is not eliminated. All four victim $\times$ **Model-based attack** interactions remain close to one (IRR 0.99–1.03), consistent with the null main effect.

Returning to **H2**: the combination of victim and attacker strategy does jointly determine disclosure risk—the **Attack target** interaction is statistically significant—but the effect is dominated by the victim’s anonymization choice. **H2** thus receives partial support: strategy pairing matters, yet strong anonymization reduces disclosure risk substantially and fairly uniformly, regardless of the attack channel.

5 Conclusion and Future Work

We conducted a meta-analysis of PWS Cup 2025 under the constraints of undisclosed algorithms and a small sample size. By encoding teams’ design decisions as categorical features, we found a pattern consistent with H1: generation- or reconstruction-based strategies were associated with higher utility and anonymization scores than direct perturbation-based approaches in this contest setting. H2 received partial support, indicating that although strategy pairing influences disclosure risk, strong anonymization remains the dominant factor. Because the analysis is based on a small observational sample reconstructed from public presentations, these results should be read as exploratory evidence rather than confirmatory conclusions. Future work will apply the proposed feature taxonomy to systematically collect strategy information in future PWS Cup competitions.

Material availability. Publicly released team materials and competition results are available at <https://www.iwsec.org/pws/2025/cup25.html>. Our reconstructed team-level feature data, codebook, and analysis code are available at https://github.com/kiti1924/PWSCUP2025_REVIEW_PUB. Materials not released by the teams cannot be redistributed.

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References

1. Allard, T., Béziaud, L., Gambs, S.: SNAKE challenge: Sanitization algorithms under attack. In: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. pp. 5010–5014. CIKM ’23 (2023). <https://doi.org/10.1145/3583780.3614754>
2. Burnham, K.P., Anderson, D.R.: Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach. Springer, New York, NY, 2 edn. (2002). <https://doi.org/10.1007/b97636>
3. Carlini, N., Chien, S., Nasr, M., Song, S., Terzis, A., Tramèr, F.: Membership inference attacks from first principles. In: 2022 IEEE Symposium on Security and Privacy (SP). pp. 1897–1914 (2022). <https://doi.org/10.1109/SP46214.2022.9833649>
4. Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. p. 785–794. KDD ’16, Association for Computing Machinery, New York, NY, USA (2016), <https://doi.org/10.1145/2939672.2939785>
5. Dwork, C., McSherry, F., Nissim, K., Smith, A.: Calibrating noise to sensitivity in private data analysis. In: Halevi, S., Rabin, T. (eds.) Theory of Cryptography. pp. 265–284. Springer Berlin Heidelberg, Berlin, Heidelberg (2006)

6. Gadotti, A., Rocher, L., Houssiau, F., Crețu, A.M., de Montjoye, Y.A.: Anonymization: The imperfect science of using data while preserving privacy. *Science Advances* **10**(29), eadn7053 (2024), <https://www.science.org/doi/abs/10.1126/sciadv.adn7053>
7. Hedges, L.V.: Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics* **6**(2), 107–128 (1981). <https://doi.org/10.3102/10769986006002107>
8. Hurvich, C.M., Tsai, C.L.: Regression and time series model selection in small samples. *Biometrika* **76**(2), 297–307 (06 1989), <https://doi.org/10.1093/biomet/76.2.297>
9. Kikuchi, H., Yamaguchi, T., Hamada, K., Yamaoka, Y., Oguri, H., Sakuma, J.: Ice and fire: Quantifying the risk of re-identification and utility in data anonymization. In: 2016 IEEE 30th International Conference on Advanced Information Networking and Applications (AINA). pp. 1035–1042 (2016). <https://doi.org/10.1109/AINA.2016.151>
10. Murakami, T., Arai, H., Hamada, K., Hatano, T., Iguchi, M., Kikuchi, H., Kuro-masa, A., Nakagawa, H., Nakamura, Y., Nishiyama, K., Nojima, R., Oguri, H., Watanabe, C., Yamada, A., Yamaguchi, T., Yamaoka, Y.: Designing a location trace anonymization contest. *Proceedings on Privacy Enhancing Technologies* **2023**(1), 225–243 (Jan 2023), <http://dx.doi.org/10.56553/popets-2023-0014>
11. Patki, N., Wedge, R., Veeramachaneni, K.: The synthetic data vault. In: 2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA). pp. 399–410 (2016). <https://doi.org/10.1109/DSAA.2016.49>
12. Shafeinejad, M., He, X., Alinoori, M., Jewell, J., Ayromlou, S., Pang, W., Chattrath, V., Sharma, G., Pandya, D.: MIDST challenge at SaTML 2025: Membership inference over diffusion-models-based synthetic tabular data (2025), <https://arxiv.org/abs/2603.19185>
13. Shokri, R., Stronati, M., Song, C., Shmatikov, V.: Membership inference attacks against machine learning models. In: 2017 IEEE Symposium on Security and Privacy (SP). pp. 3–18 (2017). <https://doi.org/10.1109/SP.2017.41>
14. Stadler, T., Oprisanu, B., Troncoso, C.: Synthetic data – anonymisation groundhog day. In: 31st USENIX Security Symposium (USENIX Security 22). pp. 1451–1468. USENIX Association, Boston, MA (Aug 2022), <https://www.usenix.org/conference/usenixsecurity22/presentation/stadler>
15. Sweeney, L.: k-anonymity: A model for protecting privacy. *International journal of uncertainty, fuzziness and knowledge-based systems* **10**(05), 557–570 (2002)
16. United Nations. Economic Commission for Europe: Synthetic data for official statistics: a starter guide. United Nations (2022)
17. Walonoski, J., Kramer, M., Nichols, J., Quina, A., Moesel, C., Hall, D., Duffett, C., Dube, K., Gallagher, T., McLachlan, S.: Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *Journal of the American Medical Informatics Association* **25**(3), 230–238 (03 2018), <https://doi.org/10.1093/jamia/ocx079>
18. Xu, L., Skoularidou, M., Cuesta-Infante, A., Veeramachaneni, K.: Modeling tabular data using conditional gan. In: Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., Garnett, R. (eds.) *Advances in Neural Information Processing Systems*. vol. 32. Curran Associates, Inc. (2019), https://proceedings.neurips.cc/paper_files/paper/2019/file/254ed7d2de3b23ab10936522dd547b78-Paper.pdf

19. Zarifzadeh, S., Liu, P., Shokri, R.: Low-cost high-power membership inference attacks. In: Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., Berkenkamp, F. (eds.) Proceedings of the 41st International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 235, pp. 58244–58282. PMLR (21–27 Jul 2024), <https://proceedings.mlr.press/v235/zarifzadeh24a.html>

A Main Results and Statistics of PWS CUP 2025

Table 1: Main results of PWS Cup 2025. U_1 – U_4 are the four utility sub-scores (basic statistics; logistic regression; Kruskal–Wallis; XGBoost); U is the utility score; $Pen.$ and $Anon.$ are the anonymization penalty and score; $Total = U + Anon.$; $Atk.$ is the attack score. Team 21 withdrew; team 10 submitted no attack.

Team	U_1	U_2	U_3	U_4	U	$Pen.$	$Anon.$	Total	Rank	Atk.
1	38.06	16.30	19.33	18.21	91.89	13	70.47	162.36	8	113.81
2	37.46	17.30	19.90	17.26	91.92	14	72.86	164.78	6	105.43
3	33.56	8.26	16.48	17.64	75.93	20	65.73	141.66	17	105.81
4	35.56	6.01	19.07	17.34	77.98	10	78.60	156.58	11	101.47
5	38.38	16.09	19.51	17.66	91.64	19	66.75	158.39	10	107.74
6	39.63	14.21	19.61	17.80	91.25	11	78.31	169.56	4	109.17
7	33.07	5.83	19.88	15.31	74.09	17	71.91	146.00	16	103.93
8	37.67	15.99	19.77	17.75	91.18	10	78.63	169.81	3	121.13
9	33.26	12.99	17.37	17.42	81.04	13	74.30	155.34	12	106.84
10	38.30	16.09	19.69	17.84	91.93	14	72.21	164.14	7	13.79
11	37.66	9.77	19.19	17.91	84.54	14	69.32	153.86	13	104.06
12	31.54	13.30	15.68	17.69	78.21	15	73.90	152.11	14	106.60
13	35.80	17.45	19.63	17.96	90.84	13	76.50	167.34	5	106.93
14	33.75	9.19	16.96	17.46	77.36	20	24.64	102.00	22	101.22
15	29.10	6.48	15.29	17.11	67.99	17	67.81	135.80	19	106.09
16	33.17	14.34	16.70	17.54	81.75	12	77.28	159.03	9	107.59
17	32.28	9.98	15.32	17.82	75.40	19	30.44	105.84	21	102.79
18	38.18	15.62	19.78	17.98	91.55	10	79.28	170.83	1	108.39
19	27.95	10.03	15.12	17.85	70.95	12	75.11	146.06	15	101.50
20	39.24	17.75	19.78	17.91	94.67	11	75.57	170.24	2	99.99
21	—	—	—	—	—	—	—	—	24	—
22	36.80	13.14	16.76	17.72	84.41	20	47.69	132.10	20	102.83
23	23.06	10.44	15.32	17.54	66.36	12	70.29	136.65	18	100.63
24	19.77	12.50	19.12	18.20	69.59	20	0.00	69.59	23	104.46

Table 2: Breakdown of the number of teams for each feature. Each entry in the second column consists of a tuple of feature values and a corresponding tuple of team counts.

Features	Number of teams for category
Anonymization	
Base	(B, others) = (13, 7)
Synthesis	(yes, no) = (10, 10)
Noise	(yes, no) = (6, 14)
Optimization	(yes, no) = (6, 14)
Post-processing	(swap, none) = (10, 10)
Learning Models	
B-base use	(yes, no) = (18, 2)
Non-B use	(yes, no) = (4, 16)
B processing	(none, synthetic, NA) = (13, 5, 2)
Attacks	
Attack target	(anon, both) = (15, 4)
Non default distance	(yes, no) = (8, 11)
Model-based attack	(yes, no) = (6, 13)